

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

WEEK 14: LECTURE NOTE

CHAPTER 5. APPLICATIONS OF INTEGRATION
Cont...

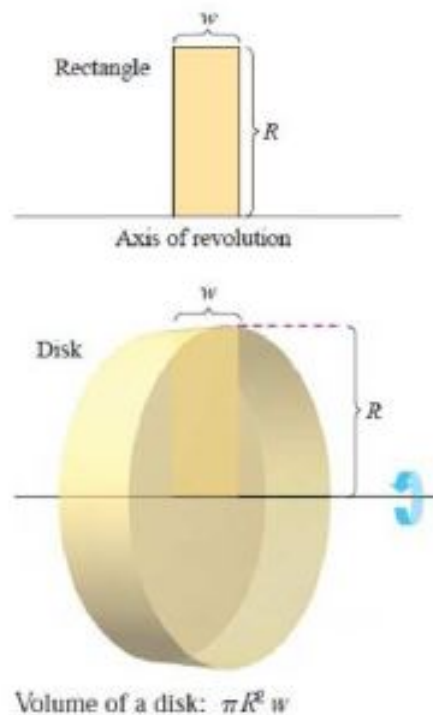
Volumes of Revolution about x -axis or y -axis

When a region in the plane is revolved about a line, the resulting solid is a **solid of revolution**, and the line is called the **axis of revolution**.

The simplest such solid is a **right circular cylinder** or **disk**, which is formed by revolving a rectangle about an axis adjacent to one side of the rectangle. The volume of such a disk is

$$\text{Volume of disk} = \text{area of disk} \times \text{width of disk} = \pi R^2 w,$$

where R is the radius of the disk and w is the width.



Proposition 1 (THE DISK AND WASHER METHOD). *The volume of the solid formed by revolving the region bounded by*

1. *the graph of $y = f(x)$ on the interval $[a, b]$ and the x -axis is given by*

$$V = \pi \int_a^b [R(x)]^2 dx = \pi \int_a^b [f(x)]^2 dx.$$

THE DISK METHOD

2. the graph of $x = g(y)$ on the interval $[c, d]$ and the y -axis is given by

$$V = \pi \int_c^d [R(y)]^2 dy = \pi \int_c^d [g(y)]^2 dy.$$

THE DISK METHOD

3. the graphs of $y = f(x)$ and $y = g(x)$ on the interval $[a, b]$, where $f(x) \geq g(x)$, about the x -axis is given by

$$V = \pi \int_a^b ([f(x)]^2 - [g(x)]^2) dx.$$

THE WASHER METHOD

4. the graphs of $x = f(y)$ and $x = g(y)$ on the interval $[c, d]$, where $f(y) \geq g(y)$, about the y -axis is given by

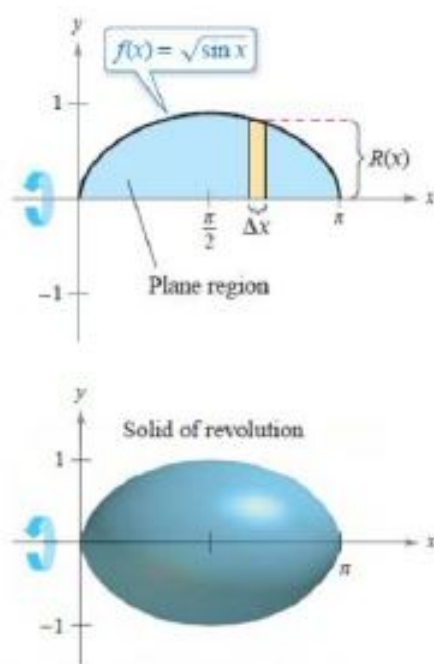
$$V = \pi \int_c^d ([f(y)]^2 - [g(y)]^2) dy.$$

THE WASHER METHOD

Example 2. Find the volume of the solid formed by revolving the region bounded by the graph of

$$f(x) = \sqrt{\sin x}, \quad 0 \leq x \leq \pi$$

and the x -axis about the x -axis.



Solution. We can see that the radius of this solid is $R(x) = f(x) = \sqrt{\sin x}$. So, the volume of the solid of revolution is

$$V = \pi \int_a^b [R(x)]^2 dx$$

Apply disk method.

$$= \pi \int_0^\pi [\sqrt{\sin x}]^2 dx$$

Substitute $\sqrt{\sin x}$ for $R(x)$.

$$= \pi \int_0^\pi \sin x dx$$

Simplify.

$$= \pi [-\cos x]_0^\pi$$

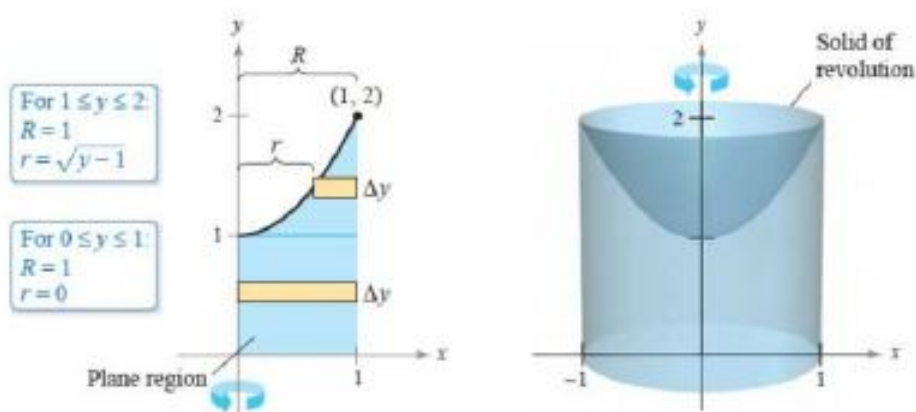
Integrate.

$$= 2\pi.$$

Example 3 (Integrating with Respect to y , Two-Integral Case). Find the volume of the solid formed by revolving the region bounded by the graphs of

$$y = x^2 + 1, \quad y = 0, \quad x = 0, \text{ and } x = 1$$

about the y -axis, as shown in Figure below.



Solution. For the region shown in the Figure above, the outer radius is simply $R = 1$. There is, however, no convenient formula that represents the inner radius. When $0 \leq y \leq 1$, $r = 0$, but when $1 \leq y \leq 2$, r is determined by the equation $y = x^2 + 1$, which implies that $r = \sqrt{y - 1}$.

$$r(y) = \begin{cases} 1, & 0 \leq y \leq 1 \\ \sqrt{y - 1}, & 1 \leq y \leq 2 \end{cases}$$

Using this definition of the inner radius, we can use two integrals to find the volume of the solid of revolution.

$$\begin{aligned} V &= \pi \int_0^1 [1^2 - 0^2] dy + \pi \int_1^2 [1^2 - (\sqrt{y - 1})^2] dy && \text{Apply Washer method.} \\ &= \pi \int_0^1 1 dy + \pi \int_1^2 (2 - y) dy && \text{Simplify.} \\ &= \pi [y]_0^1 + \pi \left[2y - \frac{y^2}{2} \right]_1^2 && \text{Integrate.} \\ &= \pi + \pi \left(4 - 2 - 2 + \frac{1}{2} \right) \\ &= \pi + \frac{\pi}{2} \\ &= \frac{3\pi}{2}. \end{aligned}$$